

Sumcheck-free hypercube inner products from table-form Kronecker encodings

Research note (companion to Kronecker-FRI)

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Abstract

We encode a multilinear polynomial by its *table*, $V_f(X) = \sum_w f(w)X^w$, instead of its coefficients. Two quantities then become single coefficients of univariate products: the evaluation $\tilde{f}(z)$, with a product-form kernel, and the hypercube inner product $\sum_w a(w)b(w)$ of two committed tables, through the reversal $V_b^*(X) = X^{N-1}V_b(1/X)$. Because a smooth domain is closed under inversion, the verifier reads V_b^* from the commitment to b as a virtual word, and the Kronecker-FRI machinery (opening polynomial, virtual word, one folding test) proves $\sum_w a(w)b(w) = S$ with no sumcheck, with soundness error $3M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$. With geometric weights γ^w and DEEP quotients as further virtual words, the same method gives a sumcheck-free Hadamard check $a \circ b = c$ for three committed tables, with soundness error $2(N - 1)/|\mathbb{F}| + 8M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$. Finally, any list of evaluations, inner products and Hadamard checks on the same commitments is proved by one folding test, with soundness error $2(N - 1)/|\mathbb{F}| + tM/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$ for $t + 1$ batched words; the term $2(N - 1)/|\mathbb{F}|$ is paid once, whatever the number of Hadamard checks. All results are machine-checked in Lean 4.

Setting. Notation and parameters are those of Kronecker-FRI (KF) [1]: $n \geq 1$, $N = 2^n$, a smooth domain $L \leq \mathbb{F}^\times$ of order $M = 2^{n+R}$ with $R \geq 2$, codes $\mathcal{C}_j$, the folding test $\Pi_{\text{FT}}[\ell, \kappa]$ with error $\varepsilon_{\text{fold}}$ (KF Thm 5.24), fibre distance Δ_0^{fib} , and $0 < \delta \leq \delta^* = (1 - \rho)/2$. A bit vector $w \in \{0, 1\}^n$ is identified with the integer $\sum_k w_k 2^{k-1} \in [0, N - 1]$.

1 Table encoding and two extraction identities

Definition 1 (Table encoding, reversal). For a table $f : \{0, 1\}^n \rightarrow \mathbb{F}$, $V_f(X) = \sum_w f(w)X^w \in \mathbb{F}[X]_{<N}$ and $\text{Enc}^T(f) = \text{ev}_L(V_f)$. For $P \in \mathbb{F}[X]_{<N}$, its reversal is $P^*(X) = X^{N-1}P(1/X)$, so $[X^j]P^* = [X^{N-1-j}]P$ and $(P^*)^* = P$.

$f \mapsto V_f$ is a linear bijection onto $\mathbb{F}[X]_{<N}$, so Enc^T is a linear bijection onto $\mathcal{C}_0$ (as KF Lemma 6.3(1)); it is the Kronecker encoding of the multilinear polynomial whose coefficients are the values of f , and needs no Möbius transform. In particular $\text{Enc}^T(f) = \text{Enc}(f)$ as maps on vectors of length N : a table-form commitment is an ordinary Kronecker-FRI commitment to the vector of values, with the same Merkle tree, and all of KF’s results on commitments (unique decoding, extraction, batching) apply to it unchanged.

Lemma 2 (Evaluation). For $z \in \mathbb{F}^n$ let $E_z^*(X) = \prod_{k=1}^n (z_k + (1 - z_k)X^{2^{k-1}})$. Then $\tilde{f}(z) = [X^{N-1}]V_f E_z^*$, where $\tilde{f}$ is the multilinear extension of f .

Proof. Expanding the product, $E_z(X) := \sum_w \text{eq}(w, z)X^w = \prod_k ((1 - z_k) + z_k X^{2^{k-1}})$, and $X^{N-1}E_z(1/X) = \prod_k X^{2^{k-1}}((1 - z_k) + z_k X^{-2^{k-1}}) = E_z^*(X)$ since $N - 1 = \sum_k 2^{k-1}$. Hence $[X^{N-1-w}]E_z^* = [X^w]E_z = \text{eq}(w, z)$ and $[X^{N-1}]V_f E_z^* = \sum_w f(w) \text{eq}(w, z) = \tilde{f}(z)$. $\square$

Lemma 3 (Inner product). *For tables a, b : $\sum_w a(w)b(w) = [X^{N-1}] V_a V_b^*$.*

Proof. $[X^{N-1}] V_a V_b^* = \sum_{i=0}^{N-1} [X^i] V_a [X^{N-1-i}] V_b^* = \sum_i [X^i] V_a [X^i] V_b = \sum_w a(w)b(w)$. $\square$

Both are instances of one fact: for $U \in \mathbb{F}[X]_{<N}$ and a public $K \in \mathbb{F}[X]_{<N}$, $[X^{N-1}] UK = \sum_i [X^i] U [X^{N-1-i}] K$. KF's split lemma (Lemma 4.10) and identity lemma (Lemma 4.12) use only $\deg U, \deg K \leq N-1$, so they hold verbatim for UK replaced by $V_a E_z^*$ or $V_a V_b^*$:

Lemma 4 (Split and identity, general form). *Let $U, K, A, H \in \mathbb{F}[X]_{<N}$ and $v \in \mathbb{F}$. (1) There are $A, H \in \mathbb{F}[X]_{<N}$ with $XUK = A + vX^N + X^{N+1}H$ iff $v = [X^{N-1}] UK$. (2) $\Phi = XUK - A - vX^N - X^{N+1}H$ has degree at most $2N$, and $\Phi = 0$ implies $v = [X^{N-1}] UK$.*

Proof. As in KF Lemmas 4.10 and 4.12: $\deg(UK) \leq 2N-2$, and the three parts of the right-hand side occupy the disjoint degree ranges $[0, N-1]$, $\{N\}$, $[N+1, 2N]$; compare the coefficients of X^N . $\square$

Lemma 5 (Reversed words). *For $w \in \mathbb{F}^L$ put $w^*(\xi) = \xi^{N-1}w(\xi^{-1})$. Then: (1) $w = \text{ev}_L(P)$ with $P \in \mathbb{F}[X]_{<N}$ implies $w^* = \text{ev}_L(P^*)$; in particular $w \mapsto w^*$ maps $\mathcal{C}_0$ onto $\mathcal{C}_0$; (2) $w \mapsto w^*$ is a linear bijection of $\mathbb{F}^L$, and $w^* = c^*$ at ξ iff $w = c$ at ξ^{-1} ; (3) a set $T \subseteq L$ is fibre-closed iff $T^{-1} = \{\xi^{-1} : \xi \in T\}$ is, and $|T^{-1}| = |T|$; (4) $h(\xi) = w^*(\xi)$ is computed from the value of w at $\xi^{-1} \in L$, and a fold check at $\pm\xi$ reads w on the fibre $\{\pm\xi^{-1}\}$.*

Proof. L is a group, so $\xi \mapsto \xi^{-1}$ is a bijection of L mapping $-\xi$ to $-\xi^{-1}$ (as $-1 \in L$); and $\xi^{N-1} \neq 0$. (1) $\xi^{N-1}P(\xi^{-1}) = P^*(\xi)$, and $P \mapsto P^*$ is a bijection of $\mathbb{F}[X]_{<N}$. (2)–(4) follow from the first sentence. $\square$

2 The inner-product protocol

Definition 6 (Π_{IP}). Instance: oracles $w_a, w_b \in \mathbb{F}^L$ and $S \in \mathbb{F}$; honestly $w_a = \text{Enc}^T(a)$, $w_b = \text{Enc}^T(b)$ and $S = \sum_w a(w)b(w)$.

1. The prover sends $w_A = \text{ev}_L(A)$, where $XV_a V_b^* = A + SX^N + X^{N+1}H$ (Lemma 4); the verifier sends $\beta \in \mathbb{F}$.
2. Put the virtual words w_b^* (Lemma 5) and $h(\xi) = (\xi w_a(\xi) w_b^*(\xi) - w_A(\xi) - S\xi^N)/\xi^{N+1}$, and

$$w_0 = w_a + \beta w_b^* + \beta^2 w_A + \beta^3 h.$$

3. Run $\Pi_{\text{FT}}[\ell, \kappa]$ (or Π_{FT}^C , KF Prop. 5.28) on w_0 .

Per query point the verifier reads $w_a(\pm\xi)$, $w_b(\pm\xi^{-1})$ and $w_A(\pm\xi)$, and does $O(1)$ extra field operations beyond KF (no kernel to evaluate).

Theorem 7 (Completeness). *If $w_a = \text{ev}_L(V_a)$, $w_b = \text{ev}_L(V_b)$ and $S = \sum_w a(w)b(w)$, the honest prover is accepted with probability 1.*

Proof. By Lemma 3 and Lemma 4(1), $A, H \in \mathbb{F}[X]_{<N}$ exist; by Lemma 5(1) $w_b^* = \text{ev}_L(V_b^*)$; evaluating the identity at ξ gives $h = \text{ev}_L(H)$. So $w_0 = \text{ev}_L(V_a + \beta V_b^* + \beta^2 A + \beta^3 H)$ with a polynomial of degree $< N$, and KF Lemma 5.9(1) applies. $\square$

Definition 8 (Relation). $\mathcal{R}_{\text{IP}}^\delta$: $((w_a, w_b; S), (a, b))$ with $\Delta_0^{\text{fib}}(w_a, \text{Enc}^T(a)) \leq \delta$, $\Delta_0^{\text{fib}}(w_b, \text{Enc}^T(b)) \leq \delta$ and $\sum_w a(w)b(w) = S$.

Lemma 9 (Batching, degree 3). *Let $\mathcal{G}$ be the set of β for which w_0 agrees with some $c \in \mathcal{C}_0$ on a fibre-closed $T \subseteq L$ with $|T| \geq (1 - \delta)M$. If $|\mathcal{G}| > 3M$, then $(w_a, w_b; S)$ has a witness in $\mathcal{R}_{\text{IP}}^\delta$.*

Proof. The words w_0 form the curve of degree 3 generated by (w_a, w_b^*, w_A, h) , and every $\beta \in \mathcal{G}$ has $\Delta(w_0, \mathcal{C}_0) \leq \delta$. By correlated agreement for curves (KF Thm 2.16, from [2], with $t = 3$) there are $P_0, P_1, P_2, P_3 \in \mathbb{F}[X]_{<N}$ and a set Agr , $|\text{Agr}| \geq (1 - \delta)M$, on which $w_a = \text{ev}(P_0)$, $w_b^* = \text{ev}(P_1)$, $w_A = \text{ev}(P_2)$, $h = \text{ev}(P_3)$ (take Agr maximal).

Value. Let $\Phi = XP_0P_1 - P_2 - SX^N - X^{N+1}P_3$. On Agr , $\Phi(\xi) = \xi w_a(\xi)w_b^*(\xi) - w_A(\xi) - S\xi^N - \xi^{N+1}h(\xi) = 0$ by the definition of h . Since $\deg \Phi \leq 2N < (1 - \delta)M \leq |\text{Agr}|$ (KF Lemma 7.2, which uses $R \geq 2$), $\Phi = 0$, and Lemma 4(2) gives $S = [X^{N-1}]P_0P_1$. Let a, b be the tables with $V_a = P_0$ and $V_b = P_1^*$; then $V_b^* = P_1$ and $S = \sum_w a(w)b(w)$ by Lemma 3.

Bad challenges. For $\xi \notin \text{Agr}$, $d_\xi(Z) = \sum_{i=0}^3 Z^i(u_i - \text{ev}(P_i))(\xi)$, with $(u_0, \dots, u_3) = (w_a, w_b^*, w_A, h)$, is a nonzero polynomial of degree ≤ 3 ; its roots over all such ξ form a set Bad with $|\text{Bad}| \leq 3\delta M < 3M < |\mathcal{G}|$. Take $\beta \in \mathcal{G} \setminus \text{Bad}$ with its c and T , and $\hat{c} = \sum_i \beta^i \text{ev}(P_i) \in \mathcal{C}_0$. Then c and $\hat{c}$ agree with w_0 on $T \cap \text{Agr}$, of size $\geq (1 - 2\delta)M \geq N$, so $c = \hat{c}$; and for $\xi \in T$, $d_\xi(\beta) = w_0(\xi) - \hat{c}(\xi) = 0$ forces $\xi \in \text{Agr}$. So $T \subseteq \text{Agr}$.

Distances. $w_a = \text{ev}(V_a)$ on the fibre-closed T , so $\Delta_0^{\text{fb}}(w_a, \text{Enc}^T(a)) \leq \delta$ (KF Lemma 5.11(2)). $w_b^* = \text{ev}(V_b^*) = \text{ev}(V_b)^*$ on T , so by Lemma 5(2) $w_b = \text{ev}(V_b)$ on T^{-1} , which is fibre-closed of the same size (Lemma 5(3)); so $\Delta_0^{\text{fb}}(w_b, \text{Enc}^T(b)) \leq \delta$. $\square$

Theorem 10 (Soundness). *If $(w_a, w_b; S)$ has no witness in $\mathcal{R}_{\text{IP}}^\delta$, every prover is accepted with probability at most*

$$\frac{3M}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa < \frac{4M}{|\mathbb{F}|} + (1 - \delta)^\kappa.$$

Proof. Identical to KF Thm 7.8, with Lemma 9 in place of KF Lemma 7.4: for $\beta \notin \mathcal{G}$ (probability $\geq 1 - 3M/|\mathbb{F}|$) and good r (probability $\geq 1 - \varepsilon_{\text{fold}}$, KF Thm 5.24(1)), the codeword chain (KF Prop. 5.23) shows that $\text{dens}_\ell(w_\ell) \geq 1 - \delta$ would put β in $\mathcal{G}$, the witness set $\mathcal{W}_\ell(w_\ell)$ being fibre-closed (KF Lemma 7.7); so the query term is at most $(1 - \delta)^\kappa$ (KF Lemma 3.3). $\square$

Round-by-round knowledge soundness, the extractor (decode w_a and w_b at radius δ^*) and binding follow as in KF Thm 7.11 and Cor. 7.12, with round-1 error $3M/|\mathbb{F}|$.

Remark 11 (One commitment, three kinds of openings). The same table-form commitment $\text{Enc}^T(f)$ supports: evaluation openings at any z (KF's protocol with the kernel E_z^* of Lemma 2, evaluated by the verifier with $O(n)$ operations); hypercube sums ($z = (\frac{1}{2}, \dots, \frac{1}{2})$ up to 2^n , or the kernel $\prod_k (1 + X^{2^{k-1}})$ directly); and inner products with any other committed table (Π_{IP}). Inner products with a *public* table p are the special case $K = V_p^*$, useful when $V_p^*(\xi)$ is cheap to evaluate (product form). All of these batch with KF's batched openings (KF Thm 11.2).

3 A sumcheck-free Hadamard check

SNARKs of Spartan/HyperPlonk type need, beyond linear functionals and inner products, the statement $a(w)b(w) = c(w)$ for all w (a *Hadamard check*), for three committed tables.

Geometric weights. For $\gamma \in \mathbb{F}$,

$$\sum_w \gamma^w a(w)b(w) = [X^{N-1}] V_a(\gamma X) V_b^*(X), \quad \sum_w \gamma^w c(w) = V_c(\gamma),$$

the first by Lemma 3 applied to the tables $w \mapsto \gamma^w a(w)$ and b . If $d = a \circ b - c \neq 0$, then $D_d(X) = \sum_w d(w)X^w$ is a nonzero polynomial of degree $\leq N - 1$, and the two sides differ unless $D_d(\gamma) = 0$. The values $V_a(\gamma\xi)$ and $V_c(\gamma)$ lie outside L ; the prover commits to $Q(X) = V_a(\gamma X)$, and both are tied to the commitments by DEEP quotients, which are virtual words like h .

Definition 12 (Quotient words). For $w \in \mathbb{F}^L$ and $y, x \in \mathbb{F}$, put $q[w, y, x](\xi) = (w(\xi) - y)/(\xi - x)$, with the value 0 at $\xi = x$ if $x \in L$.

Lemma 13 (DEEP step on an agreement set). Let $x, y \in \mathbb{F}$, $P, R \in \mathbb{F}[X]_{<N}$ and $T \subseteq L$ with $|T| > N + 1$. If $w = \text{ev}(P)$ and $q[w, y, x] = \text{ev}(R)$ on T , then $P(x) = y$ and $R = (P - y)/(X - x)$. Conversely, if $x \notin L$, $w = \text{ev}_L(P)$ and $P(x) = y$, then $q[w, y, x] = \text{ev}_L((P - y)/(X - x))$, a polynomial of degree $\leq N - 2$.

Proof. $(X - x)R - P + y$ has degree $\leq N$ and vanishes on $T \setminus \{x\}$, which has more than N points, so it is 0; evaluate at x . The converse is the factor theorem. $\square$

Definition 14 (Π_{Had}). Instance: oracles $w_a, w_b, w_c \in \mathbb{F}^L$; honestly $w_f = \text{Enc}^T(f)$ for $f \in \{a, b, c\}$ and $a \circ b = c$.

1. The verifier sends γ , uniform in $\mathbb{F}$. The prover sends $w_Q = \text{ev}_L(Q)$, where $Q(X) = V_a(\gamma X)$.
2. The verifier sends θ , uniform in $\mathbb{F}$. The prover sends $y_1 = V_a(\gamma\theta)$, $y_3 = V_c(\gamma)$ and $w_A = \text{ev}_L(A)$, where $XQV_b^* = A + y_3X^N + X^{N+1}H$ (Lemma 4).
3. The verifier sends β , uniform in $\mathbb{F}$. With $h(\xi) = (\xi w_Q(\xi) w_b^*(\xi) - w_A(\xi) - y_3\xi^N)/\xi^{N+1}$ and

$$(u_0, \dots, u_8) = (w_a, w_c, w_Q, w_b^*, w_A, h, q[w_a, y_1, \gamma\theta], q[w_Q, y_1, \theta], q[w_c, y_3, \gamma]),$$

$$\text{put } w_0 = \sum_{i=0}^8 \beta^i u_i.$$

4. Run $\Pi_{\text{FT}}[\ell, \kappa]$ (or Π_{FT}^C) on w_0 .

The challenges are unrestricted: soundness holds for all values, and a pole $\gamma\theta, \theta, \gamma$ in L affects only completeness. Using y_1 for both $V_a(\gamma\theta)$ and $Q(\theta)$ is the consistency check $V_a(\gamma\theta) = Q(\theta)$. Per query point the verifier reads w_a, w_c, w_Q, w_A on the coset of ξ and w_b on the coset of ξ^{-1} , and does $O(1)$ field operations and three inversions per point (one batched inversion per query) beyond the folding test.

Theorem 15 (Completeness). If $w_f = \text{Enc}^T(f)$ for $f \in \{a, b, c\}$ and $a \circ b = c$, the honest prover is accepted whenever $\gamma\theta, \theta, \gamma \notin L$, hence with probability at least $1 - 3M/|\mathbb{F}|$.

Proof. $Q \in \mathbb{F}[X]_{<N}$; $y_1 = V_a(\gamma\theta) = Q(\theta)$; and $y_3 = V_c(\gamma) = \sum_w \gamma^w c(w) = \sum_w \gamma^w a(w)b(w) = [X^{N-1}]QV_b^*$, so A, H exist and $h = \text{ev}_L(H)$ as in Π_{IP} . By Lemma 13 the three quotient words are codewords when the poles lie outside L , and then w_0 is the codeword of a polynomial of degree $< N$. Each of the events $\gamma \in L$, $\theta \in L$, $\gamma\theta \in L$ has probability at most $M/|\mathbb{F}|$ (for $\gamma \neq 0$, at most M values of θ put $\gamma\theta$ in L ; for $\gamma = 0$, none). $\square$

Definition 16 (Relation). $\mathcal{R}_{\text{Had}}^\delta$: $((w_a, w_b, w_c), (a, b, c))$ with $\Delta_0^{\text{fib}}(w_f, \text{Enc}^T(f)) \leq \delta$ for $f \in \{a, b, c\}$ and $a \circ b = c$.

Lemma 17 (Batching, degree 8). Fix $\gamma, \theta \in \mathbb{F}$ and the prover's w_Q, y_1, y_3, w_A . Let $\mathcal{G}$ be the set of β for which w_0 agrees with some $c' \in \mathcal{C}_0$ on a fibre-closed $T \subseteq L$ with $|T| \geq (1 - \delta)M$. If $|\mathcal{G}| > 8M$, there are tables a, b, c and $\hat{Q} \in \mathbb{F}[X]_{<N}$ with

1. $\Delta_0^{\text{fib}}(w_f, \text{Enc}^T(f)) \leq \delta$ for $f \in \{a, b, c\}$, and $\Delta_0^{\text{fib}}(w_Q, \text{ev}_L(\hat{Q})) \leq \delta$;
2. $V_a(\gamma\theta) = y_1 = \hat{Q}(\theta)$;
3. $V_c(\gamma) = y_3 = [X^{N-1}] \hat{Q} V_b^*$.

Proof. Correlated agreement on one fibre-closed set (the step shared with Lemma 9; Lean: `curve_agreement`, $t = 8$) gives $P_0, \dots, P_8 \in \mathbb{F}[X]_{<N}$ and a fibre-closed T , $|T| \geq (1 - \delta)M > 2N \geq N + 2$ (KF Lemma 7.2), with $u_i = \text{ev}(P_i)$ on T . As in Lemma 9, $X P_2 P_3 - P_4 - y_3 X^N - X^{N+1} P_5$ vanishes on T and has degree $\leq 2N$, so $y_3 = [X^{N-1}] P_2 P_3$. Lemma 13 with $(P_0, P_6, \gamma\theta)$, (P_2, P_7, θ) and (P_1, P_8, γ) gives $P_0(\gamma\theta) = y_1$, $P_2(\theta) = y_1$ and $P_1(\gamma) = y_3$. Let $V_a = P_0$, $V_c = P_1$, $V_b = P_3^*$ and $\hat{Q} = P_2$; then (2) and (3) hold, and (1) follows from agreement on T (for w_b , on T^{-1} , Lemma 5). $\square$

Theorem 18 (Soundness). *Let $R \geq 2$ and $0 < \delta \leq \delta^*$. If (w_a, w_b, w_c) has no witness in $\mathcal{R}_{\text{Had}}^\delta$, every prover is accepted with probability at most*

$$\varepsilon_{\text{Had}} = \frac{2(N-1)}{|\mathbb{F}|} + \frac{8M}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1-\delta)^\kappa.$$

Proof. For $\delta \leq \delta^*$ each word has at most one codeword within fibre distance δ (KF Lemma 6.3(2)), so the tables and $\hat{Q}$ of Lemma 17 are the decodings of w_a, w_b, w_c, w_Q . It suffices to consider deterministic provers (a randomised prover is an average of deterministic ones), so w_Q is a function of γ and the round-2 messages are functions of (γ, θ) . Call (γ, θ) *good* if, for every choice of the prover's round-2 messages, $|\mathcal{G}| \leq 8M$. For good (γ, θ) the argument of KF Thm 7.8 applied to the rest of the protocol bounds the acceptance probability by $8M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1-\delta)^\kappa$. It remains to bound the probability that (γ, θ) is not good.

Case 1: some w_f has no table within fibre distance δ . Lemma 17(1) fails, so every (γ, θ) is good.

Case 2: w_a, w_b, w_c decode to $\hat{a}, \hat{b}, \hat{c}$ with $\hat{a} \circ \hat{b} \neq \hat{c}$. Let $D = D_{\hat{a} \circ \hat{b} - \hat{c}} \neq 0$. The event $D(\gamma) = 0$ has probability at most $(N-1)/|\mathbb{F}|$. Fix γ with $D(\gamma) \neq 0$, and the prover's w_Q (a function of γ). If w_Q has no codeword within fibre distance δ , (γ, θ) is good for every θ by Lemma 17(1). If it decodes to $\hat{Q} = V_{\hat{a}}(\gamma X)$, then Lemma 17(3) would give $\sum_w \gamma^w \hat{c}(w) = V_{\hat{c}}(\gamma) = [X^{N-1}] V_{\hat{a}}(\gamma X) V_b^* = \sum_w \gamma^w \hat{a}(w) \hat{b}(w)$, i.e. $D(\gamma) = 0$; so every θ is good. Otherwise $\hat{Q} - V_{\hat{a}}(\gamma X)$ is a nonzero polynomial of degree $\leq N-1$, fixed before θ is drawn; the event $\hat{Q}(\theta) = V_{\hat{a}}(\gamma \theta)$ has probability at most $(N-1)/|\mathbb{F}|$, and outside it Lemma 17(2) fails, so θ is good. $\square$

So $\varepsilon_{\text{Had}} < (8M + 2N)/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1-\delta)^\kappa$, against $3M/|\mathbb{F}|$ for Π_{IP} and $2M/|\mathbb{F}|$ for an evaluation. The curve has degree 8 because w_a and w_c must be in the batch themselves: a quotient word alone determines w_a on T only as a polynomial of degree $\leq N$, which does not identify the decoding.

Knowledge follows: if the acceptance probability exceeds ε_{Had} , the instance has a witness, and the extractor decodes w_a, w_b, w_c at radius δ (KF Cor. 7.12).

Remark 19 (Consequence). Linear functionals (evaluations, with the kernel E_z^*), inner products (Π_{IP}) and Hadamard checks (Π_{Had}) are all proved on table-form Kronecker-FRI commitments by one folding test each, with no sumcheck and a verifier that is an FRI verifier plus $O(1)$ operations per query point (plus $O(n)$ for E_z^*). For R1CS-type SNARKs what remains is the sparse matrix-vector products Az, Bz, Cz , which Spartan handles with its own sparse-polynomial commitment; whether those admit a coefficient-extraction form is the next question.

4 One folding test for a batch of statements

A SNARK needs many statements at once on the same commitments: Hadamard checks, inner products and evaluations. Since every statement of Sections 1–3 is checked by words in one curve, all of them can share one curve and one folding test.

Definition 20 (Statements). Let $w_1, \dots, w_r \in \mathbb{F}^L$ be commitments. A *statement* is one of the following, with indices in $\{1, \dots, r\}$:

- **Eval**(i, z, v) with $z \in \mathbb{F}^n$, $v \in \mathbb{F}$: the table f_i of w_i has $\tilde{f}_i(z) = v$;
- **IP**(i, i', S) with $S \in \mathbb{F}$: $\sum_w f_i(w) f_{i'}(w) = S$;
- **Had**(i, i', k): $f_i \circ f_{i'} = f_k$.

For a list $\Sigma = (\sigma_1, \dots, \sigma_J)$ of $J \geq 1$ statements, let D be the set of commitment indices read directly (i in **Eval** and **IP**; i, k in **Had**), B the set of indices read reversed (i' in **IP** and **Had**), and $\mathcal{H} \subseteq \{1, \dots, J\}$ the positions of the Hadamard checks. A table may play several roles. The relation

$\mathcal{R}_{\text{Batch}}^\delta$ holds for $((w_i)_i, \Sigma)$ and tables $(f_i)_{i \in D \cup B}$ if $\Delta_0^{\text{fib}}(w_i, \text{Enc}^T(f_i)) \leq \delta$ for all $i \in D \cup B$ and every statement of Σ holds for these tables.

For each statement σ_j the protocol uses a triple (U_j, K_j, v_j) , with $U_j, K_j \in \mathbb{F}[X]_{<N}$, and the identity $XU_jK_j = A_j + v_jX^N + X^{N+1}H_j$ of Lemma 4: (V_{f_i}, E_z^*, v) for **Eval** (i, z, v) (Lemma 2); $(V_{f_i}, V_{f_{i'}}^*, S)$ for **IP** (i, i', S) (Lemma 3); $(Q_j, V_{f_{i'}}^*, y_3^{(j)})$ with $Q_j(X) = V_{f_i}(\gamma X)$ for $\sigma_j = \text{Had}(i, i', k)$ (Section 3). The verifier reads $U_j(\xi)$ and $K_j(\xi)$ as $w_i(\xi)$ and $E_z^*(\xi)$, as $w_i(\xi)$ and $w_{i'}^*(\xi)$, or as $w_Q^{(j)}(\xi)$ and $w_{i'}^*(\xi)$.

Definition 21 (Π_{Batch}). Instance: $(w_i)_i$ and Σ .

1. The verifier sends γ , uniform in $\mathbb{F}$. For each $j \in \mathcal{H}$ the prover sends $w_Q^{(j)} = \text{ev}_L(Q_j)$.
2. The verifier sends θ , uniform in $\mathbb{F}$. For each $j \in \mathcal{H}$, with $\sigma_j = \text{Had}(i, i', k)$, the prover sends $y_1^{(j)} = V_{f_i}(\gamma\theta)$ and $y_3^{(j)} = V_{f_k}(\gamma)$; for each $j \leq J$ it sends $w_A^{(j)} = \text{ev}_L(A_j)$.
3. The verifier sends β , uniform in $\mathbb{F}$. The words are, in a fixed order: w_i for $i \in D$; w_i^* for $i \in B$; $w_Q^{(j)}$ for $j \in \mathcal{H}$; $w_A^{(j)}$ and $h_j(\xi) = (\xi U_j(\xi)K_j(\xi) - w_A^{(j)}(\xi) - v_j\xi^N)/\xi^{N+1}$ for $j \leq J$; and $q[w_i, y_1^{(j)}, \gamma\theta]$, $q[w_Q^{(j)}, y_1^{(j)}, \theta]$, $q[w_k, y_3^{(j)}, \gamma]$ for $j \in \mathcal{H}$, $\sigma_j = \text{Had}(i, i', k)$. They are $t+1 = |D| + |B| + 2J + 4|\mathcal{H}|$ words $u_0, \dots, u_t$; put $w_0 = \sum_{s=0}^t \beta^s u_s$.
4. Run $\Pi_{\text{FT}}[\ell, \kappa]$ (or Π_{FT}^C) on w_0 .

γ and θ are shared by all Hadamard checks. All $w_Q^{(j)}$ can be committed in one Merkle tree whose leaves hold one value of each, and likewise all $w_A^{(j)}$; a query then opens, beyond the folding test, one path per commitment in D , one per commitment in B (on the inverse coset), and at most two paths for the prover's oracles (one if $\mathcal{H} = \emptyset$), whatever J .

Theorem 22 (Completeness). *If $w_i = \text{Enc}^T(f_i)$ and every statement holds, the honest prover is accepted whenever $\mathcal{H} = \emptyset$ or $\gamma\theta, \theta, \gamma \notin L$; so with probability at least $1 - 3M/|\mathbb{F}|$.*

Proof. Each $w_A^{(j)}$, h_j and quotient word is a codeword, by the completeness proofs of Sections 2 and 3 and Lemma 2 for **Eval** (where E_z^* is a public polynomial of degree $< N$). $\square$

Lemma 23 (Batching, degree t). *Fix γ, θ and the prover's messages, and let $\mathcal{G}$ be the set of β for which w_0 agrees with some $c' \in \mathcal{C}_0$ on a fibre-closed $T \subseteq L$ with $|T| \geq (1 - \delta)M$. Let $R \geq 2$ and $0 < \delta \leq \delta^*$. If $|\mathcal{G}| > tM$, there are tables $(f_i)_{i \in D \cup B}$ and polynomials $\hat{Q}_j \in \mathbb{F}[X]_{<N}$ ($j \in \mathcal{H}$) such that $\Delta_0^{\text{fib}}(w_i, \text{Enc}^T(f_i)) \leq \delta$, $\Delta_0^{\text{fib}}(w_Q^{(j)}, \text{ev}_L(\hat{Q}_j)) \leq \delta$, and*

1. *every Eval and IP statement of Σ holds for the f_i ;*
2. *for every $j \in \mathcal{H}$ with $\sigma_j = \text{Had}(i, i', k)$: $V_{f_i}(\gamma\theta) = y_1^{(j)} = \hat{Q}_j(\theta)$ and $V_{f_k}(\gamma) = y_3^{(j)} = [X^{N-1}] \hat{Q}_j V_{f_{i'}}^*$.*

Proof. Correlated agreement on one fibre-closed set (**curve.agreement** with this t) gives polynomials of degree $< N$ for all the words and a fibre-closed T with $|T| \geq (1 - \delta)M > 2N$ on which all words agree with them. Tables: V_{f_i} is the polynomial of w_i for $i \in D$; for $i \in B$, V_{f_i} is the reversal of the polynomial of w_i^* , and w_i agrees with $\text{Enc}^T(f_i)$ on T^{-1} (Lemma 5). If $i \in D \cap B$, both give tables within fibre distance δ of w_i , which coincide by unique decoding. For each j , the identity of h_j on T and Lemma 4(2) give $v_j = [X^{N-1}] U_j K_j$ with the polynomials of the words; for **Eval** this is $v = \tilde{f}_i(z)$ by Lemma 2, for **IP** it is $S = \sum_w f_i(w) f_j(w)$ by Lemma 3, and for **Had** it is the second equality of (2) with $\hat{Q}_j$ the polynomial of $w_Q^{(j)}$. The three quotient words of each $j \in \mathcal{H}$ give the other equalities of (2) by Lemma 13, as in Lemma 17. $\square$

Theorem 24 (Soundness). *Let $R \geq 2$ and $0 < \delta \leq \delta^*$. If $((w_i)_i, \Sigma)$ has no witness in $\mathcal{R}_{\text{Batch}}^\delta$, every prover is accepted with probability at most*

$$\varepsilon_{\text{Batch}} = \frac{2(N-1)}{|\mathbb{F}|} + \frac{tM}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1-\delta)^\kappa.$$

Proof. As for Theorem 18: call (γ, θ) good if $|\mathcal{G}| \leq tM$ for all round-2 messages; for good (γ, θ) the acceptance probability is at most $tM/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1-\delta)^\kappa$. If some w_i , $i \in D \cup B$, has no table within fibre distance δ , every pair is good by Lemma 23. Otherwise let $\hat{f}_i$ be the decodings; some statement σ_{j^*} is false for them, and j^* depends only on the instance. If σ_{j^*} is an Eval or IP statement, Lemma 23(1) fails, so every pair is good. If $\sigma_{j^*} = \text{Had}(i, i', k)$, the proof of Theorem 18 applies verbatim to this one check, with Lemma 23(2) for j^* in place of Lemma 17: the pair is good unless $D_{\hat{f}_i \circ \hat{f}_{i'} - \hat{f}_k}(\gamma) = 0$, or $\hat{Q}_{j^*} \neq V_{\hat{f}_i}(\gamma X)$ and $\hat{Q}_{j^*}(\theta) = V_{\hat{f}_i}(\gamma \theta)$, events of probability at most $(N-1)/|\mathbb{F}|$ each. $\square$

When $\mathcal{H} = \emptyset$ the same proof shows that every pair is good, so the first term can be omitted (this refinement is not in the Lean formalisation). Only one false statement is needed in the proof, so the term $2(N-1)/|\mathbb{F}|$ from γ, θ is paid once, not once per Hadamard check; the term $tM/|\mathbb{F}|$ grows linearly in the number of words, by $4M/|\mathbb{F}|$ per Hadamard check and $2M/|\mathbb{F}|$ per other statement. For $|\mathbb{F}| = p^2 \approx 2^{128}$, $M = 2^{22}$ and $t = 100$, this term is about 2^{-99} . Knowledge follows as before: if the acceptance probability exceeds $\varepsilon_{\text{Batch}}$, the instance has a witness, obtained by decoding each w_i .

n	one batched proof ($t+1=16$)			four separate proofs		
	prove (ms)	verify (ms)	proof (KiB)	prove (ms)	verify (ms)	proof (KiB)
12	38.2	7.21	397.9	54.3	13.85	795.4
14	165.2	8.20	479.5	224.9	16.17	1017.2
16	716.2	8.90	587.5	959.8	18.70	1315.3
18	3154.6	10.22	701.2	4116.7	22.59	1658.7
20	13187.4	11.32	804.5	18012.6	25.43	1957.9

Table 1: The statements $\text{Had}(a, b, c)$, $\text{IP}(a, d, S)$, $\text{Eval}(a, z_1, v_1)$, $\text{Eval}(c, z_2, v_2)$ on four table-form commitments, proved by one Π_{Batch} proof or by four (each a Π_{Batch} proof of one statement). Configuration KF-8, 148 queries, 32-byte salts, $\mathbb{F}_{p^2}$; one core; medians. Source: `bench/batch.csv`, `cargo run --release --example batch_bench`. The prototype prover is not optimised.

One proof for the four statements is 2.0 to 2.4 times smaller and 1.9 to 2.2 times faster to verify than four proofs (2.4 and 2.2 at $n=20$), and its prover takes 23 to 30% less time.

5 Relation to known techniques and status

Writing an inner product as the middle coefficient of a product with a reversed polynomial is classical, and appears in pairing-based multilinear commitments (Mercury [3] uses the table form with KZG). DEEP quotients for out-of-domain values are standard in FRI-based STARKs. The new points here are: (1) the reversed virtual word w_b^* , which reads the second vector from the *same* Reed–Solomon commitment thanks to $L^{-1} = L$; (2) the reuse of Kronecker-FRI’s opening polynomial, virtual word and single folding test, with a fully proved soundness bound (Lemma 9); (3) the table-form evaluation kernel E_z^* , which gives evaluation openings on the same commitments; (4) a Hadamard check in which the out-of-domain values are DEEP quotient words inside the same batch as the opening polynomial, so that one degree-8 curve and one folding test carry the whole argument (Lemma 17).

Prover cost and the sumcheck baseline. The prover of Π_{IP} computes one product $V_a V_b^*$ of two polynomials of degree $< N$ over the base field (three NTTs of size $2N$), one NTT of size M and one Merkle tree for w_A , and then runs the folding test; every round-1 object lies in the base field, since V_a, V_b do. The natural baseline is the sumcheck route: n rounds of a degree-2 sumcheck on $\sum_w a(w)b(w)$, followed by openings of a and b at the random point. For that baseline the fastest prover we know of in our own work is the tree circuit of [4], which produces the evaluations, hypercube sums and sumcheck round polynomials of a multilinear polynomial from one binary-tree recursion; the comparison of the two routes at equal soundness is left to the implementation paper.

Status.

- *Lean 4.* Sections 1–2 are formalised in the Kronecker-FRI Lean project, files `TableForm.lean` and `InnerProduct.lean`, without `sorry` and with the single axiom of that project (correlated agreement for curves, KF Thm 2.16, here with $t = 3$). Lemma 2: `eval_table`; Lemma 3: `inner_product`; Lemma 4: `splitG_iff`, `natDegree_PhiG_le`, `identityG`; Lemma 5: `wrev_ev`, `wrev_eq_iff`, `inv_image_closed`, `ncard_inv_image`; completeness: `honestIP_prob_one`; Lemma 9: `batching_IP`; soundness: `soundness_IP`, and `soundnessC_IP` for any folding arity 2^k ; knowledge and binding: `knowledge_IP`, `binding_IP`. The common step of KF Lemma 7.4 and Lemma 9, correlated agreement on one fibre-closed set for a curve of any degree, is the separate theorem `curve_agreement`.
- *Rust.* The module `ip` of the Kronecker-FRI crate implements Π_{IP} with Merkle caps, arity 2^k and Fiat–Shamir, on the commitments of `pcs`. Its tests check Lemmas 3, 4 and 5(1), completeness for $n \leq 10$, two rates and arities 2 to 16, and the rejection of three cheating provers (a wrong S ; a wrong S absorbed into a polynomial A of degree N , as in KF Remark 4.13; opening polynomials computed from a table $b' \neq b$).
- *Lean 4, Section 3.* File `Hadamard.lean`, same axiom, no `sorry`. Lemma 13: `deep_step`, `deep_honest`; completeness: `honestHad_accepts`, and `honestHad_prob` for the bound $1 - 3M/|\mathbb{F}|$; Lemma 17: `batching_Had`; Theorem 18: `prob_bad_le` (the bad pairs (γ, θ)), `soundness_Had`, and `soundnessC_Had` for any folding arity 2^k ; knowledge: `knowledge_Had`. The step “for good (γ, θ) , the argument of KF Thm 7.8” is the separate theorem `ft_family_bound`: for any family of words $w_0(\beta)$, the acceptance probability is at most $|\mathcal{G}|/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$.
- *Lean 4, Section 4.* File `BatchStmts.lean`, same axiom, no `sorry`. Statements: `BStmt` (`eval`, `ip`, `had`), `Dset`, `Bset`, `RelBatchS`; the words are indexed by the finite type `BWord` ($D \oplus B \oplus \mathcal{H} \oplus [J] \oplus [J] \oplus \mathcal{H} \times [3]$) with a fixed enumeration, and `curve_agreement_idx` is correlated agreement for such an index type. Lemma 23: `batching_B`; Theorem 24: `prob_bad_had` (one false Hadamard check), `prob_bad_B`, `soundness_B`, `soundnessC_B` (any folding arity); knowledge: `knowledge_B`; completeness: `bwords_honest`, `honestB_accepts`, `honestB_prob`.
- *Rust, Section 4.* The module `batch` implements Π_{Batch} with one multi-value Merkle tree for all w_Q and one for all w_A ; its tests check Lemma 2 for E_z^* , completeness for mixed batches (two Hadamard checks sharing a table, an inner product, two evaluations) for $n \leq 10$ and arities 2 to 16, and the rejection of a batch with one false evaluation, one false inner product, or one false Hadamard check (also against a prover that sets y_3 to the weighted inner product, which only the quotient word $q[w_c, y_3, \gamma]$ catches).
- *Rust, Section 3.* The module `had` implements Π_{Had} on the shared folding-test code of module `ft`, drawing γ, θ by rejection sampling outside the base field $\mathbb{F}_p \supseteq L$, so that no pole lies in L and completeness is perfect (the two first terms of ε_{Had} become $(N - 1)/(|\mathbb{F}| - p)$ each); its tests check completeness for $n \leq 10$, two rates and arities 2 to 16, and the rejection of a wrong y_1 , of a wrong Q (with $y_1 = Q(\theta)$), and of a false instance (c differing from $a \circ b$ in one

entry) with three choices of y_3 . Removing $q[w_c, y_3, \gamma]$ from the batch makes the false instance accepted, and a wrong pole in the verifier breaks completeness, so the tests exercise each part.

n	prove (ms)			verify (ms)			proof (KiB)		
	KF	IP	Had	KF	IP	Had	KF	IP	Had
12	7.5	6.2	17.6	2.09	2.13	4.31	138.1	171.4	289.1
14	32.2	25.9	74.0	2.56	2.60	4.97	182.3	233.4	371.1
16	141.4	109.2	314.7	3.24	3.25	5.63	249.8	308.1	458.8
18	548.2	437.8	1353.6	3.87	4.01	6.40	325.0	395.3	565.7
20	2736.6	1947.4	6004.5	4.59	5.13	7.39	391.7	471.1	662.0

Table 2: KF: one evaluation opening (Π_{KF}); IP: one inner product (Π_{IP}); Had: one Hadamard check (Π_{Had}). Configuration KF-8 of the Kronecker-FRI paper (rate 1/4, arity 8, caps of 128 nodes, $\ell = n - 8$), 148 queries, 32-byte salts, $\mathbb{F}_{p^2}$ over Goldilocks; one core of an Intel Xeon at 2.10 GHz; medians; commitments not included. Source: `bench/ip.csv`, `cargo run --release --example ip.bench`.

An inner product of two committed tables costs less prover time than one evaluation opening (the round-1 work is in the base field), about the same verifier time, and about 20% more proof, the extra openings of w_b on the inverse cosets. A Hadamard check costs about 2.2 evaluation openings of prover time (its polynomials Q , A , H lie in the extension field) and 1.7 times the proof, since five oracles are opened per query instead of two; when a , b , c are committed together, one tree with three values per leaf removes two of the five authentication paths per query.

Next. The sparse products Az , Bz , Cz .

References

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